



HarvestWise

Voice-first autonomous restocking for India's street merchants.

WhatsApp / Soundbox

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Sarvam Indic Voice AI

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Cognee Knowledge Graph

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n8n Orchestrator

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Mandi Wholesaler

Lakshmi sells vegetables in Bengaluru. Paytm made her a **merchant** — nothing made her a **restocker**. HarvestWise listens, predicts spoilage, recalculates her order and dispatches it to her mandi wholesaler — **without asking her to type a single word.**

Paytm made them merchants. Nobody made them restockers.

India's largest merchant network is also its least-served growth market: payments are digital, but the highest-stakes daily decision “what to buy at 4:30 AM” still runs on guesswork.

30M+

MERCHANTS IN PAYTM'S ECOSYSTEM

The long tail: kiranas, carts, stalls “the backbone of India's daily commerce.

1.57 Cr

MERCHANT DEVICE SUBSCRIPTIONS (SOUNDBOX / POS)

A proven billing rail merchants already pay on “and the anchor for cross-sell.

8 Cr

MONTHLY TRANSACTING USERS

Demand-side scale that makes every merchant insight immediately actionable.

The locked layer of value

Paytm sees the transactions “but the merchant's most consequential decision, **what to stock tomorrow at what quantity**, happens blind at the mandi gate. Paytm's own AI (ARMS) optimizes merchant lifecycle *internally*; the merchant's own P&L still runs on intuition. That missing merchant-facing layer is where the next phase of Paytm's growth lives.

Why now: Indic voice AI (Sarvam), agent orchestration (n8n) and knowledge-graph memory (Cognee) matured together for the first time “making a copilot for a non-English, non-dashboard user finally buildable.

Her inventory rots. Her margin goes with it.

"I'm at Kalasipalya mandi by 4:30 AM with **â,13,500 in my pocket**. I buy tomatoes, greens and beans on one heuristic: what sold yesterday. At 3 PM an unpredicted thunderstorm empties the street. By 9 PM I'm dumping **â,1400 of rotting greens** in the garbage. **I earn on Paytm. I restock on guesswork.**"

â€” LAKSHMI â€” VEGETABLE CART â€” BASAVANAGUDI, BENGALURU

âšš **Modeled across Bengaluru's 12,000+ vegetable vendors: ~â,175 Cr/yr in spoilage losses** (â,1400 ã— 12,000 ã— 365) â€” losses like these push vendors into informal high-interest debt (*meter baddi*). Baseline to be validated in pilot.

ðŸŸ¶ 25â€”35% daily spoilage

Perishable stock bought at 4:30 AM on intuition, sold across a 12-hour day. Evening leftovers go out at distress prices â€” **what doesn't sell is a 100% cash loss.**

ðŸŹ² 4:30 AM <â€”> 3 PM disconnect

Procurement decisions ignore the signals that actually move demand: **afternoon rain, micro-festivals, local UPI purchase velocity, mandi price swings** â€” all available, none connected.

ðŸ”µ Tools not built for her

Dashboards need literacy, dry hands and downtime she doesn't have. Chatbots only talk. **Wet hands, continuous customers, vernacular-only** â€” the interface must be speech.

â,13,500 / day

AVG. WORKING CAPITAL

ã—

~30%

UNSOLD AT CLOSE

âˆ’

evening salvage

DISTRESS DISCOUNTING

=

â,1400 / day

NET LOSS â€” âˆ’â,12,000/MO

Up to **30% of household take-home** erased
â€” illustrative model, spoilage band
25â€”35%

A restocking loop that speaks, acts & remembers

1 • OBSERVE

𑀓𑀲𑀭𑀺𑀓𑀺𑀲𑀺𑀓

Watches her world – hourly UPI sales velocity (Paytm signals, with consent), IMD rain probability, festival calendar, mandi rates, and **perishable decay tables** per commodity.

2 • PREDICT

𑀓𑀲𑀭𑀺𑀓𑀺𑀲𑀺𑀓𑀺𑀲𑀺𑀓

Forecasts spoilage & demand in her words: "Rain at 3 PM – coriander shelf-life drops to 4 hours; your rainy-day coriander sales fall 40%."

No new app. No login. No literacy barrier.

The interface is **voice on WhatsApp** – plus the morning briefing **played on her Paytm Soundbox**. The Soundbox already sits on her cart, already speaks her language. We upgrade it from payment confirmation to operational assistant.

3 • ACT

𑀓𑀲𑀭𑀺𑀓𑀺𑀲𑀺𑀓𑀺𑀲𑀺𑀓

Recalculates and dispatches – resizes her order (10 – 6 bundles), confirms with her by voice, sends the revised buy-order to her mandi wholesaler on WhatsApp.

4 • LEARN

𑀓𑀲𑀭𑀺𑀓𑀺𑀲𑀺𑀓𑀺𑀲𑀺𑀓

Remembers her business – which recalcs worked, which didn't, and why. Next week's order starts from this week's evidence, not zero.

Not another chatbot

Chatbots **advise** ("stock less"). HarvestWise **finishes the job** – it recalculates the order and dispatches it to the wholesaler, with a human voice-approval gate before anything that touches money.

One loop, every morning: Observe – Predict – Act – Learn.

Spoilage prevented before it's bought, orders dispatched before the mandi gate, outcomes measured the same evening – compounding daily into her take-home and Paytm's GMV.

The product story “the finale demo compresses it to one voice command” one explained recalculation “one dispatched order” one remembered outcome.

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8:30

PM • IT LEARNS

Auto-sent by the 6:30 AM n8n workflow – voice on WhatsApp **and played on her Soundbox**. 30 seconds, zero taps.

Order noted. One flag: rain at 3 PM – coriander lasts only 4 hours in humidity, and your rainy-day coriander sales drop ~40%.

Sarvam ASR â†’ intent + entities â†’ Cognee raises the spoilage flag against her history.

à²¹à³⊆à²¹à³□, à²•à²³à³□à²¹à²¿à²,à³□. ("Yes, send it.")

Dispatched: buy-order sent to the wholesaler, stock sheet updated
 gate: nothing moves without her "yes".

Logged to her Cognee memory graph: **"Rain recalc â€" worked, â‚‚1,940 kept."** Next storm, she doesn't have to think.

Built on the organizers' own stack – one job each

Experience

WHERE THE MERCHANT LIVES

WhatsApp Business API

voice notes, in & out

Paytm Soundbox

morning audio briefing

Business with Paytm

optional dashboard view

No new app, no login, no typing. Voice is the entire UI; the Soundbox is the loudspeaker.

Intelligence

SARVAM AI – INDIC SPEECH LAYER

Sarvam Speech-to-Text

ASR – 22+ Indic languages

Sarvam LLM

intent + entities + reasoning

Sarvam Text-to-Speech

natural voice – 11 languages

Indic-native understanding – not an English product with translation bolted on.

Decision Core

COGNEE – CAUSAL KNOWLEDGE GRAPHS

Merchant memory graph

her history – what worked

Commodity decay tables

shelf-life – humidity – rain

Cold-start patterns

anonymized cross-merchant

Shelf-life – rain – her UPI velocity – the causal layer behind every recalc.

Orchestration

N8N – EVENT-DRIVEN ACTIONS

n8n 6:30 AM workflow

briefing – WhatsApp + Soundbox

n8n order dispatcher

revised buy-order – wholesaler

n8n approval gates

no money action without "yes"

Event-driven dispatch to mandi wholesalers; escalation & fallback built in.

Data

CONSENT-FIRST SIGNALS

Paytm merchant signals

hourly UPI velocity – consent

IMD weather API

afternoon rain probability

Mandi rate feeds

public daily price feeds

Demo runs on Paytm-transaction-shaped seeded data (labeled as such); production stays consent-first inside Paytm's rails.

– GUARDRAILS ARE THE PRODUCT

DPDP-aligned – federated learning (patterns shared, raw data never) – human approval on every money action – every recalc explained & undoable – full audit trail

Advice is cheap. Closed loops are rare.

Every alternative stops at a screen or a sentence. HarvestWise is the only one that speaks her language, recalculates her order, dispatches it, and remembers her business.

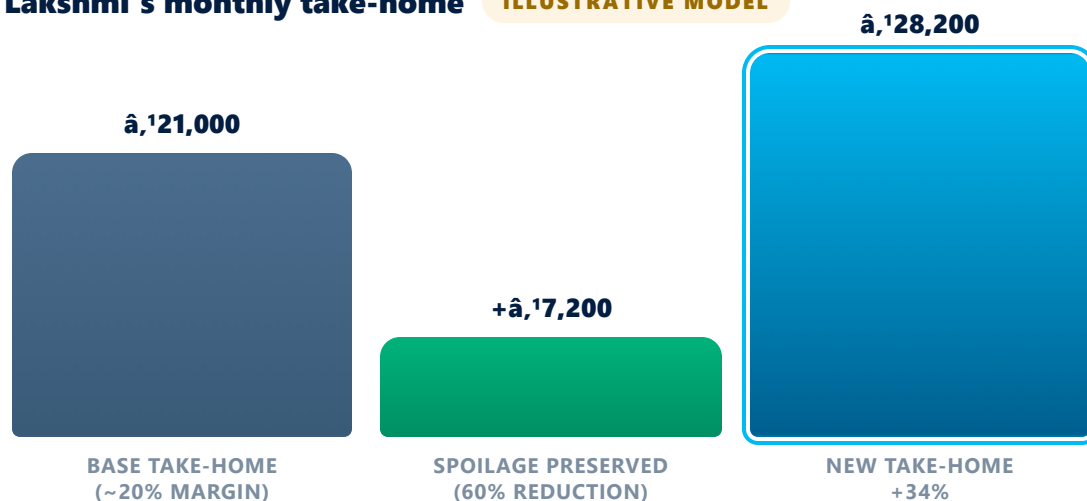
Evaluation	Traditional merchant tools	Generic AI chatbots	HarvestWise
Input modality	English forms & dashboards â€™ needs literacy, dry hands, downtime	Text-only chat â€™ assumes she can type	Zero-touch Indic voice notes on WhatsApp (Sarvam) â€™ she talks; that's the whole UI
Decision intelligence	Static inventory tallying â€™ backwards-looking	Generic public advice â€™ no context, no memory	Causal knowledge graph: shelf-life Æ— rain Æ— her hourly UPI velocity (Cognee)
Workflow completion	Manual execution â€™ she still does the work	Advises only â€™ "stock less", then stops	Closed-loop procurement: recalculated buy-order dispatched to her mandi wholesaler (n8n)
Hardware tie-in	None	None	Morning audio briefing on Paytm Soundbox â€™ payment speaker upgraded to operational assistant

MOAT

Per-merchant memory graph (Cognee) Æ— vernacular closed loop (Sarvam + n8n) Æ— Soundbox/Paytm-native distribution â€™ a compounding data advantage every single day.

Waste is her biggest expense. We give 60% of it back.

Lakshmi's monthly take-home

ILLUSTRATIVE MODEL

Assumes ₹13,500 avg. daily working capital — 30 selling days, ~20% gross margin, ₹1,400/day net spoilage loss after evening salvage. The 60% spoilage-reduction target (₹1,240/day or ₹17,200/mo) is the pilot metric we commit to validating — not a claim.

What Paytm wins

- Soundbox upgraded:** from passive payment-confirmation speaker to **active operational assistant** — a deeper, stickier device relationship.
- GMV up:** better-stocked merchants capture demand they currently miss — target **+8% GMV per merchant**.
- Underwriting data moat:** proprietary inventory & procurement telemetry derisks **Paytm Merchant Loans** — replacing "meter baddi" with formal credit.
- Mission delivered:** financial inclusion that shows up as a healthier family P&L — the story of Build for India.

₹17,200PRESERVED / VENDOR /
MO***₹1,105 Cr+**WORKING CAPITAL / YR
— 12K VENDORS***₹1,175 Cr+**CURRENT ANNUAL
SPOILAGE LOSS (CITY-
WIDE EST.)

*Model: 12,000 vendors — ₹1,240/day — 365 — ₹450 directional; validated in pilot. Impact framed as money preserved, not vanity metrics.

Billed where she already pays. Free where she starts.

Seed

â‚¹10 / forever

- **Daily demand alerts** â€” weather + spoilage risk, one voice note + Soundbox briefing
- Yesterday's sales recap
- Builds the daily habit â€” **zero CAC** via Paytm for Business & Soundbox base

Pro

â‚¹149 / month

- **Autonomous restocking:** order recalculation + dispatch to her wholesaler
- Price-drop supplier alerts & end-of-day **P&L voice audit**
- Proposed billing: **settlement-linked micro-deduction** â€” subject to Paytm integration approval; **simulated in our prototype**. No invoice, no friction.

SETTLEMENT-BILLED

Platform Revenues

Take-rate on value created

- **Supplier marketplace:** % on executed mandi orders
- **Credit distribution:** procurement telemetry derisks Paytm Merchant Loans
- **Anonymized demand curves** (DPDP-compliant) for supply-chain aggregators & mandi logistics planning

Why merchants buy

â‚¹149/mo vs â‚¹7,200/mo preserved â€” pays for itself in one saved evening of stock.

Why it compounds

Every day improves her memory graph â€” advice quality compounds, churn falls.

Unit economics

Zero-CAC Paytm-native distribution & 80%+ SaaS margin & data flywheel into lending.

Why Paytm wins

The deepest merchant lock-in since the Soundbox: the device confirmed payments; we grow them.

THE WEDGE & THE PLATFORM Start: perishable carts & stalls (highest pain, clearest signal) & kirana fresh categories & 30M+ merchants as the daily restocking layer of the Paytm ecosystem.

From this deck to a pilot

NOW · FINALE

MVP on partner stack

Synthetic 90-day sales data; 3 n8n workflows (briefing, order recalc + dispatch, day-wrap learning); Sarvam voice loop; Cognee memory. Live demo, one phone.

MONTHS 1 · 3

500-merchant pilot

KR Market, Russell Market & Madivala with vendor associations + Paytm's merchant network. Validate the 60% spoilage-reduction metric.

MONTHS 4 · 6

Paytm-native integration

API partnership with Paytm for Business; Soundbox briefing rail; mandi logistics channels wired into n8n dispatch.

BEYOND

The restocking layer

Perishables · kirana fresh · all categories. Procurement telemetry · merchant credit. Every merchant, a copilot.

"Every morning at 4:30, India's merchants guess. From tomorrow, they guess **with a copilot.**"

TEAM

Sairam · Venkata Narayana G V · Malaravan P

BUILT WITH

Sarvam AI · n8n · Cognee

ASK

A 500-merchant pilot with Paytm's network

Solve with AI. Build for India.